

Loop Depth	Reference Loop
0..	for (int n = 0; n < 1; n += 1)
1...	for (int h = 0; h < 32; h += 1)
2....	for (int w = 0; w < 32; w += 1)
	for (int ci = 0; ci < 3; w += 1)
	// other statement;
3.....	for (int co = 0; co < 64; co += 1)
	out[n][co][h][w] += ... ;

Loop Depth	Target loop
0.	for (int n = 0; n < 1; n += 1)
1..	for (int wo = 0; wo < 16; wo += 1)
2..	for (int ho = 0; ho < 16; ho += 1)
3....	for (int hk = 0; hk < 2; hk += 1)
4.....	for (int co = 0; co < 64; co += 1)
5.....	for (int wk = 0; wk < 2; wk += 1)
	... = out[n][co][ho*2+hk][wo*2+wk];

Loop Depth	Transformed Target loop
0.	for (int n = 0; n < 1; n += 1)
1..	for (int ho = 0; ho < 16; ho += 1)
2...	for (int hk = 0; hk < 2; hk += 1)
3....	for (int wo = 0; wo < 16; wo += 1)
4.....	for (int wk = 0; wk < 2; wk += 1)
5.....	for (int co = 0; co < 64; co += 1)
	... = out[n][co][ho*2+hk][wo*2+wk];

Permute

Step 1. Dim-Depth Mapping

Reference	(n, co, h, w) → (0, 3, 1, 2)
Target	(n, co, {ho, hk}, {wo, wk}) → (0, 4, {2, 3}, {1, 5})

Step 2: Depth Alignment

Reference	#Depth: 4 → #Depth: 6
	(n, co, h, w) → (n, co, {h1, h2}, {w1, w2})
	(0, 3, 1, 2) → (0, 5, {1, 2}, {3, 4})

depth update

tiling

Step 3: Depth-Depth Mapping

	n, co, {h1, h2}, {w1, w2}
Reference	Depth: 0 5 1 2 3 4
Target	Depth: 0 4 2 3 1 5
	n, co, {ho, hk}, {wo, wk}

Step 4: Loop Permutation

Target	for (...) → for (...)
	for (...) → for (...)
	for (...) → for (...)
	for (...) → for (...)
	for (...) → for (...)

Permute the nesting order

1

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